AP Precalculus **Unit 1, Lesson 1.12**

Guided Notes

Name: ___________________________ Date: ________________ Period: _________

Primary Objective

By the end of this lesson, I will be able to...

1. Describe how $g(x) = af(x+h)+k$ transforms the graph of f using _________________ and _________________. *(LO 1.12.A)*
2. Determine the ______________ and ___________ of a transformed function from the parent's domain and range. *(LO 1.12.A)*

Vocabulary & Key Concepts

parent function:

transformation:

vertical translation:

horizontal translation:

vertical dilation:

horizontal dilation:

reflection:

Hook

One function — four disguises. Can you un-transform it back?

Each transformation below produces the *same* parent shape in a different location or size. Your job today is to decode the rule so you can construct *any* disguise or reverse-engineer it from a table.

Type 1 — Vertical Translation: $g(x) = f(x) + k$

Rule (EK 1.12.A.1): Adding k to the output ____________________ the graph by k units.

- $k > 0$: shift ________ $k < 0$: shift ________

Example: Let $f(x) = x^2$. Let $g(x) = f(x) + 3 = x^2 + 3$.

x	$f(x) = x^2$	$g(x) = f(x) + 3$
-2	4	________
-1	1	________
0	0	________
1	1	________
2	4	________

Effect on domain/range: Domain _____________. Range shifts _____________.

Type 2 — Horizontal Translation: $g(x) = f(x + h)$

Rule (EK 1.12.A.2): Replacing x with $x + h$ shifts the graph __________________ by $|h|$ units.

- $h > 0$: shift _________ $h < 0$: shift _________

Caution: $g(x) = f(x + 3)$ shifts the graph _________ (not right), because we need $x + 3 = 0$, so $x =$ _______.

Example: $f(x) = x^2$, $g(x) = f(x - 2) = (x - 2)^2$.

x	$f(x) = x^2$	$g(x) = f(x - 2)$
0	0	________
1	1	________
2	4	________
3	9	________
4	16	________

Effect on domain/range: Domain shifts _____________. Range _____________.

Type 3 — Vertical Dilation: $g(x) = af(x)$, $a \neq 0$

Rule (EK 1.12.A.3): Multiplying the *output* by a scales the graph _____________ by $|a|$.

- $|a| > 1$: _____________ (stretches) $0 < |a| < 1$: _____________ (compresses)
- $a < 0$: also reflects over the _____________

Example: $f(x) = x^2$; $g(x) = 2f(x) = 2x^2$; $h(x) = -f(x) = -x^2$.

x	$f(x) = x^2$	$g(x) = 2f(x)$	$h(x) = -f(x)$
-2	4	_______	_______
-1	1	_______	_______
0	0	_______	_______
1	1	_______	_______
2	4	_______	_______

Effect on domain/range: Domain _____________. Range is multiplied by _________.

Type 4 — Horizontal Dilation: $g(x) = f(bx)$, $b \neq 0$

Rule (EK 1.12.A.4): Multiplying the *input* by b scales the graph ________________ by $\frac{1}{|b|}$.

- $|b| > 1$: ________________ (compresses horizontally) $0 < |b| < 1$: ________________ (stretches horizontally)
- $b < 0$: also reflects over the ________________

Example: $f(x) = x^2$; $g(x) = f(2x) = (2x)^2 = 4x^2$.

x	$f(x) = x^2$	$g(x) = f(2x)$
-1	1	__________
$-\frac{1}{2}$	$\frac{1}{4}$	__________
0	0	__________
$\frac{1}{2}$	$\frac{1}{4}$	__________
1	1	__________

Combined Transformations: $g(x) = af(x + h) + k$

Rule (EK 1.12.A.5): Apply transformations in this order:

Step 1: Horizontal dilation/reflection (b) **Step 2:** Horizontal shift (h)
Step 3: Vertical dilation/reflection (a) **Step 4:** Vertical shift (k)

Example: Describe all transformations applied to f to produce $g(x) = -2f(x + 1) - 3$.

Parameter	**Transformation**
$a = -2$	______________________________________
$h = 1$	______________________________________
$k = -3$	______________________________________

Effect on domain/range (EK 1.12.A.6): If the domain of f is $[-2, 4]$ and the range of f is $[0, 9]$, then:

Domain of g: ____________________________ Range of g: ____________________________